Sidharth Rony, Ph.D.

United Nations University - MERIT, Boschstraat 24, 6211 AX Maastricht

Education

2018 – 2024 **PhD in Economics** University of London, Royal Holloway, United Kingdom.
Research Interests: *Skills, Technology and Labour, Machine Learning*.

2019 – 2020 **Doctoral Coursework in Economics of Education** University of Zurich, Switzerland.

2014 – 2016 **MSc Economics and Financial Research,** Maastricht University, Netherlands.
Thesis: *Signalling Asymmetric Information in Auctions.*

2007 – 2011 **B.Tech. Electronics and Communication Engineering** Amrita Vishwa Vidyapeetham, India.
Thesis: *VLSI Implementation of Neural Network With Genetic Algorithm Learning.*

Employment

Research

01.2025 – **Researcher,** United Nations University - MERIT, Maastricht University, Netherlands.
Apply advanced analytics (*network analysis, machine learning, text analysis, LLMs*) to assess how digital and green technologies affect skills demand, employment, and wage inequality in the EU.

11.2022 – 12.2024 **Research Assistant,** London School of Economics (LSE), United Kingdom.
Assisted the policy evaluation team at the What Works Centre for Local Economic Growth at the Centre for Economic Performance, LSE.

07.2022 – 10.2022 **Research Assistant,** University of London, Royal Holloway, United Kingdom.
Assisted Dr. Ija Trapeznikova in research on public-sector employment and private job creation in Ghana.

07.2021 – 10.2021 **Consultant,** Organisation for Economic Co-operation and Development (OECD), France.
Analysed and interpreted job posting data from Lightcast for multiple OECD projects.

05.2021 – 07.2021 **Research Assistant,** University of London, Royal Holloway, United Kingdom.
Collected online job advertisement data in the UK and applied NLP for text extraction.

01.2018 – 04.2018 **Research Associate,** Centre for Development Studies, Thiruvananthapuram, India.
Led a 14-member team collecting data for the Kerala Migration Survey 2018.

01.2016 – 08.2016 **Research Assistant,** Maastricht University, Netherlands.
Assisted Dr. Christian Seel in research on auction theory; responsibilities included literature classification, proofreading, and data cleaning.

Teaching

2025 *Natural Language Processing for Social Scientists (PhD, Python)*, Lecturer, UNU-MERIT.

2024 – 2025 *Family Economics (UG)*, TA, Royal Holloway, University of London.

2019 – 2025 *Econometrics (UG, R)*, TA, Royal Holloway, University of London.

2021 – 2022 *Coding and Programming for Economists (UG, Python)*, TA, Royal Holloway, University of London.

Employment (continued)

	Data Skills for Economists (UG, R), TA, Royal Holloway, University of London.
2018 – 2020	*Applied Economics and Policy (UG)*, TA, Royal Holloway, University of London.

Industry

07.2011 – 06.2014 **Electronics Engineer,** Wipro Ltd., Bangalore/Cochin, India.
C developer in the Product Engineering Service (PES) Department; designed and developed embedded systems for multiple clients.

Research

Peer Reviewed Articles

– **S. Rony**, *LLM Meets Job Advertisements: Unmasking Skill Premia in the UK*, Forthcoming in *Research in Labor Economics*, Vol. 54, Emerald Publishing. (JMP; UNU-MERIT Working Paper 2026-010; also available on SSRN), 2026. URL: https://unu-merit.nl/publications/wppdf/2026/wp2026-010.pdf.

Working Papers

– **S. Rony** and A. Badort, *How Regions Use Large Language Models*, (upon request), 2026.

– **S. Rony**, T. Ciarli, and Ö. Nomaler, *Green and Digital Technology: Exposure, Substitution and Complementarity*, (upon request), 2026.

– **S. Rony** and T. Ciarli, *Silicon Survey Agents: Replicating Expert Technology Assessments with Large Language Models*, (upon request), 2025.

– **S. Rony** and J. Patman, *Standard Occupation Classifier - A Natural Language Processing Approach*, arXiv:2511.23057, 2025. URL: https://arxiv.org/abs/2511.23057.

– **S. Rony**, *Automation and Human Capital Investment*, SSRN Working Paper No. 5821482, 2024. URL: https://ssrn.com/abstract=5821482.

Work in Progress

– A. Badort, B. Caldarola, T. Ciarli, and **S. Rony**, *Matching Lightcast Skills to the ESCO Framework*, 2026.

– A. Badort, B. Caldarola, T. Ciarli, and **S. Rony**, *Recombining Capabilities: Skill Demand Portfolios and Regional Productivity*, 2026.

– A. Badort, B. Caldarola, T. Ciarli, and **S. Rony**, *Twin Transition Skill Demand and Regional Productivity*, 2026.

– **S. Rony** and A. Badort, *Skill Relatedness and the Absorption of AI Across US Labor Markets*, 2026.

– **S. Rony** and T. Ciarli, *Green and Digital Technology: Labour Market Effects*, 2026.

– **S. Rony**, T. Ciarli, A. Marzucchi, U. Rizzo, and A. Vanegas, *The Twin Transition in Labour Demand: Green and Digital Skills in European Online Job Vacancies*, 2026.

Technical Reports

– Data Study Group Team, *Data Study Group Final Report: Defence Science & Security Laboratory (DSTL) Using Topic Modelling to Discover New Trends in the Scientific Literature*, 2023. DOI: 10.5281/zenodo.10033237.

Conference Proceedings

– V. Vijayakumar, S. Menon, K. Divakaran, and **S. Rony**, *Modified Genetic Algorithm with Controlled Population Dynamics for Efficient Resource Utilization*, Bhubaneswar, India, 2011.

Skills

Languages	English, Malayalam, Hindi, Tamil
Coding	Python, Stata, R, SQL, Matlab, C, C++

Awards and Grants

2022-2024 **Adzuna Job Advertisement Data** – Awarded data to act as lead researcher analysing COVID-19's impact on UK skills demand using Adzuna job advertisement data.

2018–2024 **Economic and Social Research Council (ESRC) Studentship** – PhD scholarship with Professor Arnaud Chevalier on "Ageing, Automatization and the Future of Skills in the Labour Market," in partnership with the OECD.

2020 **Royal Economic Society Small Academic Expenses Grant** – Awarded to attend the Doctoral Course at the University of Zurich.

2019 **RHUL Travel Grant** – Departmental funding awarded to attend the Doctoral Course at the University of Zurich.

SeNSS Research and Training Grant – Combination of SeNSS Research and Training Grant and SeNSS Summer School bursaries awarded to attend the Essex Summer School 2019.

2012 **PES–Shining Star,** Wipro Technologies, India – Awarded for outstanding performance in post-silicon validation.

Certification

2021–2026 **Office for National Statistics (ONS) Accredited Researcher.**

2023 **Associate Fellow of the Higher Education Academy**

References

Prof. Arnaud Chevalier
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Professor of Economics
Royal Holloway, University of London
Arnaud.Chevalier@rhul.ac.uk

Prof. Melanie Lührmann
PhD Supervisor
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Dr. Stijn Broecke
PhD Supervisor (OECD partnership)
Senior Economist, Employment, Labour and Social Affairs
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Dr. Tommaso Ciarli
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Senior Researcher
UNU-MERIT, Maastricht University
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